CS435 Introduction to Big Data

Unit 1. Week 1-A

Introduction

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Topics that we will discuss today

- Introduction to Big Data
- Course overview

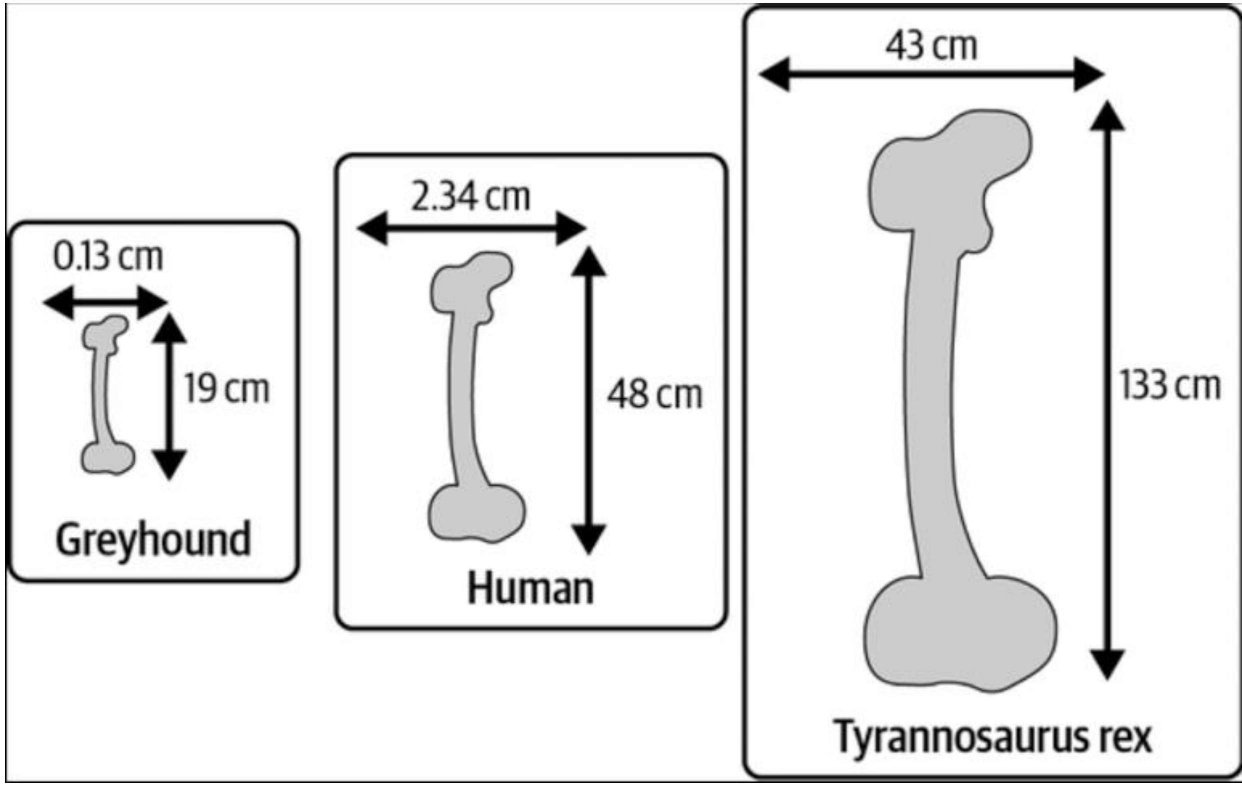
What is Big Data?

"You cannot build a giant ship
simply by enlarging a small one;
as size increases, weight grows faster than strength."

– Galileo Galilei, *Two New Sciences*, 1638

Galileo’s early scaling law research

- Femur of human is on average
  - 48cm in length
  - 2.34cm in thickness
- Femur of Greyhound
  - 19cm in length
    - >2 times smaller than human
  - 0.13cm in thickness
    - >18 times smaller than human
- How about T-Rex?



Scaling is neither free or unlimited!

Square-cube law

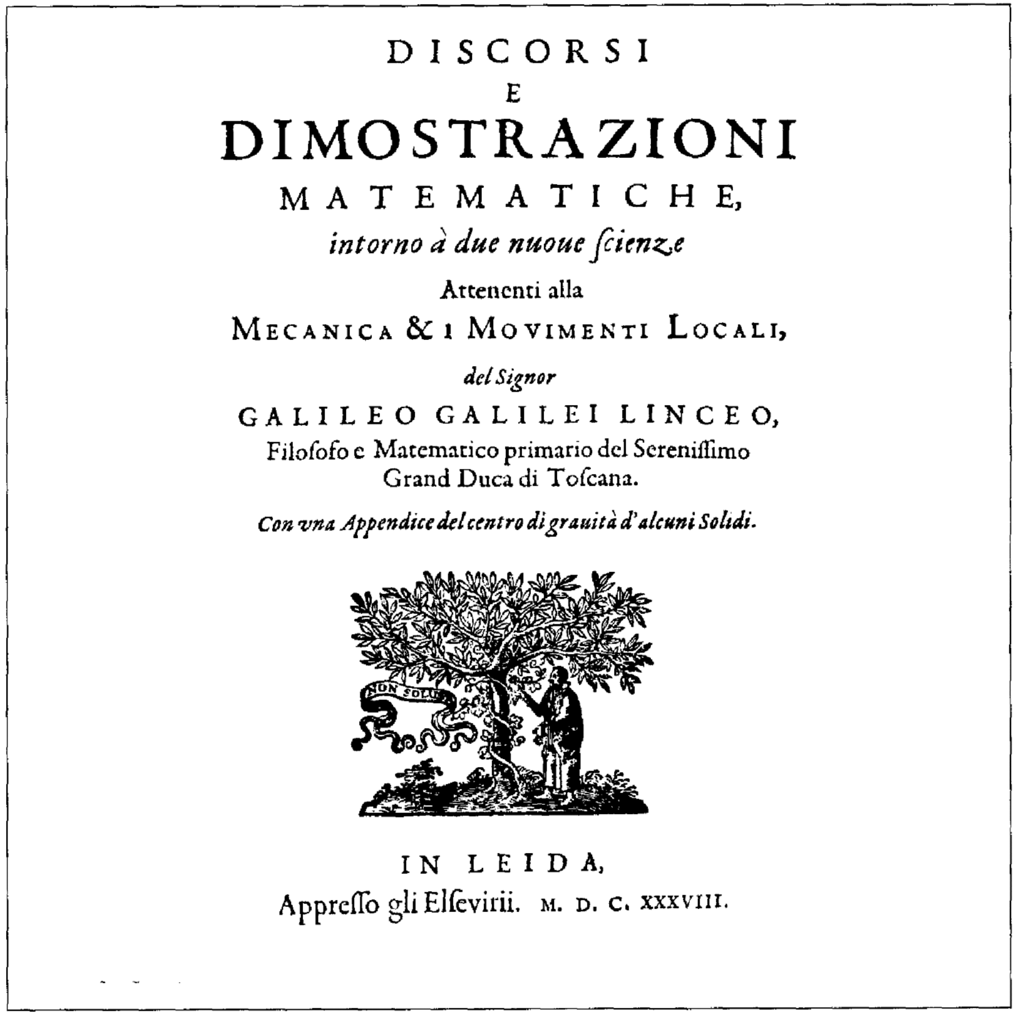
- As a shape grows in size, its volume grows faster than its surface area
- When an object undergoes a proportional increase in size, its new surface area is proportional to the square of the multiplier and its new volume is proportional to the cube of the multiplier.

$$A_2 = A_1 \left(\frac{\ell_2}{\ell_1}\right)^2$$

  - where A_1 is the original surface area and A_2 is the new area

$$V_2 = V_1 \left(\frac{\ell_2}{\ell_1}\right)^3$$

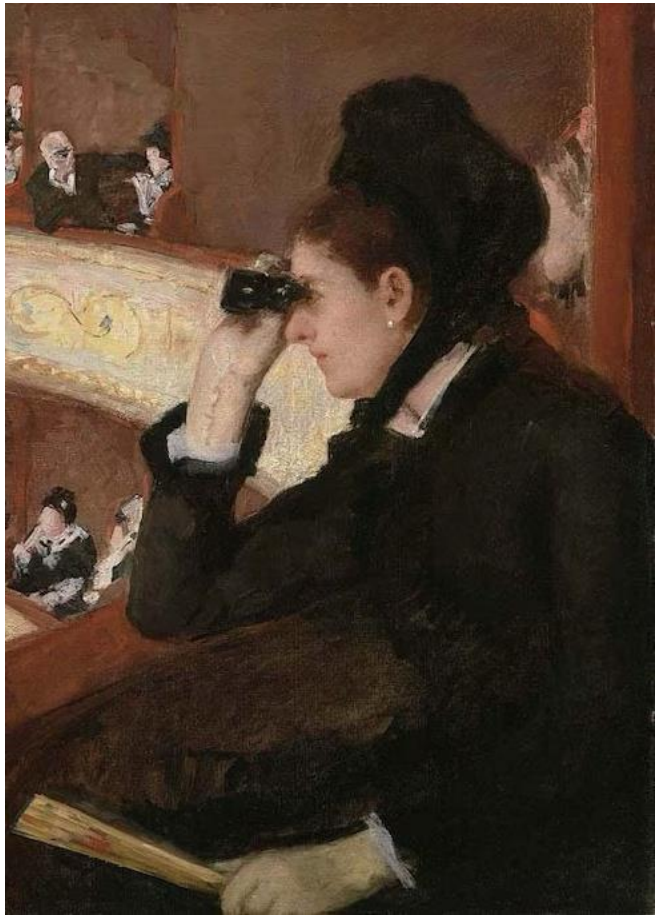
  - where V_1 is the original volume and V_2 is the new volume and l_1 is the old length and l_2 is the new length


DISCORSI
E
DIMOSTRAZIONI
MATEMATICHE,
intorno à due nuoue scienze
Attenenti alla
MECANICA & I MOVIMENTI LOCALI,
del Signor
GALILEO GALILEI LINCEO,
Filosofo e Matematico primario del Sereniſſimo
Grand Duca di Toſcana.
Con vna Appendice del centro di grauità d'alcuni Solidi.

IN LEIDA,
Appreſſo gli Elſevirii. M. D. C. XXXVIII.

The square–cube law was first mentioned in *Two New Sciences* (1638)

Big Ship vs. Toy Ship analogy In Computer Science



- Computing systems scalability
  - Scale-up solutions
    - e.g., Super computing cluster, CPU vs. GPU
  - Scale-out solutions
    - e.g., Data centers, distributed storage systems
- Algorithmic scalability
  - Algorithms more suitable for scalable computing environment
    - e.g., Merge sort vs. Tera sort
    - e.g., Scalability in Artificial Intelligence

“Big Data”

- The term “big data” refers to data that is so large, fast or complex that it’s difficult or impossible to process using traditional methods
- Big Data is about analytics of huge quantities of data in order to infer probabilities
  - With scalable algorithms and systems
- 3Vs
  - Volume
  - Velocity
  - Variety

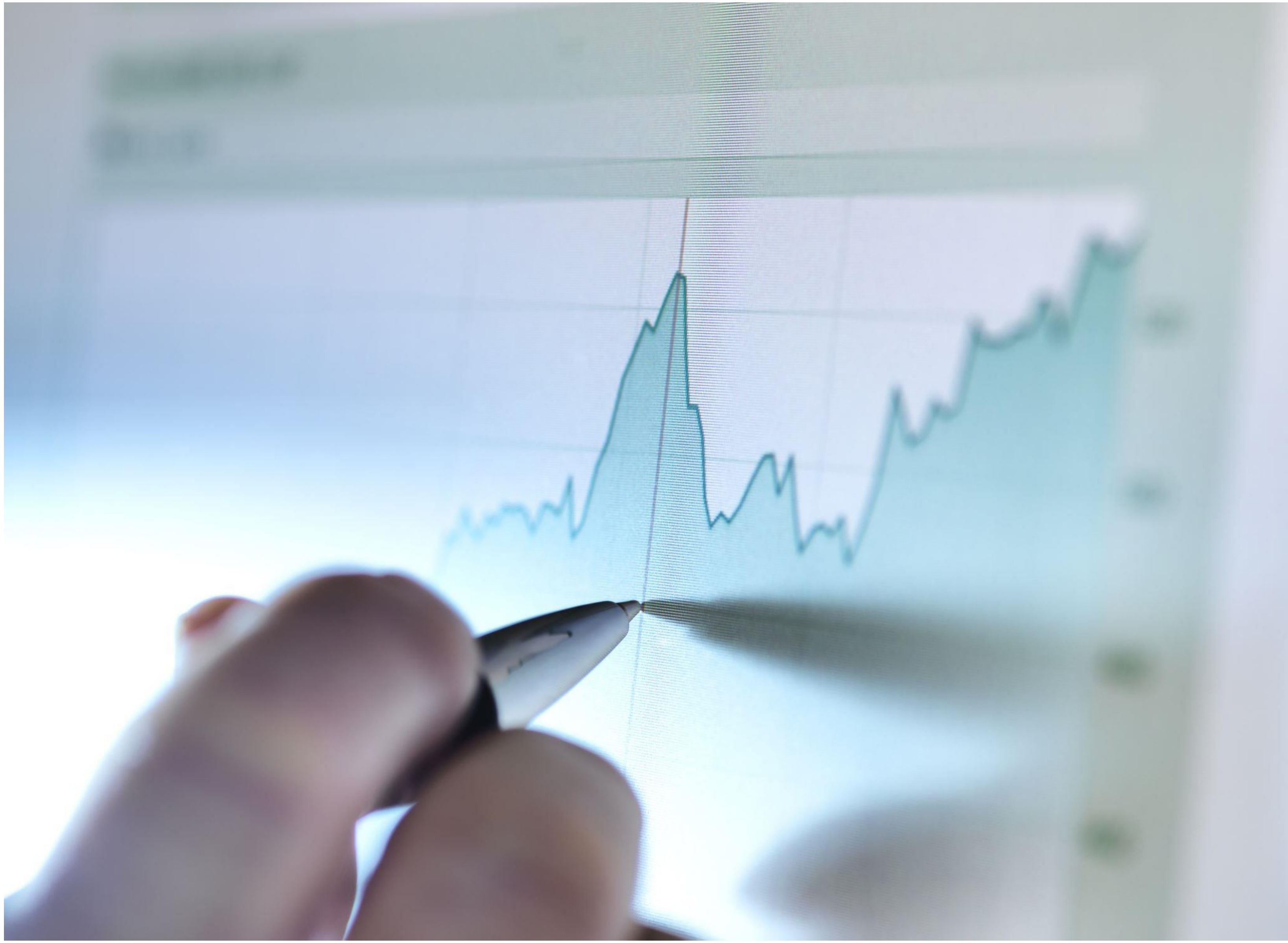
Who uses Big Data?

UBER with their Big Data

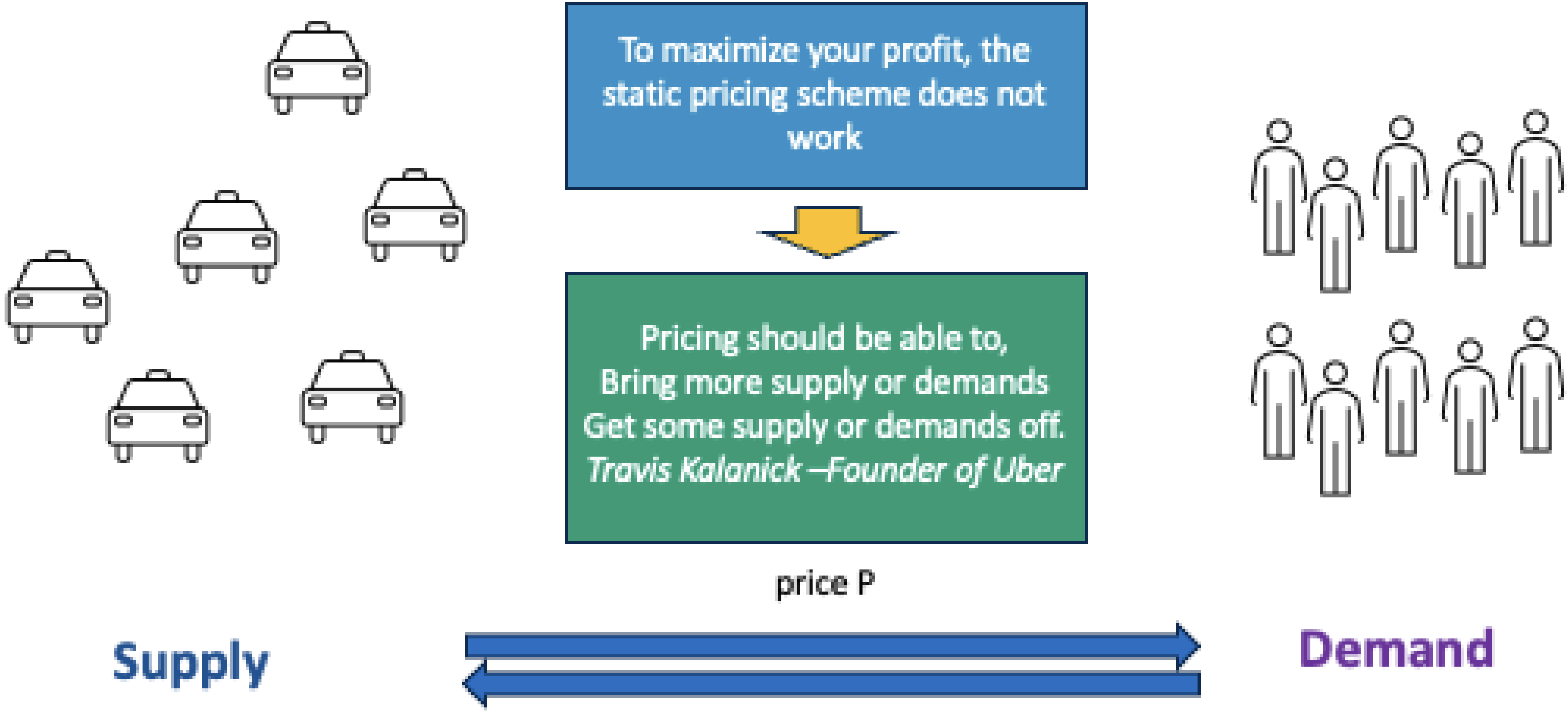
- **Data Collection**
  - Location data
  - Account/profile information
  - Demographic data
  - App usage data
  - Trip data
  - Contact data
  - Driver-related information
- **How the data is used**
  - Service enhancement
  - personalized service
  - Pricing and matching
  - Fraud detection

Pricing Your Rides

- Base rate
  - Time and distance of a trip
- Booking fee
- Busy time and area
  - When there are more riders than available drivers
  - Prices will increase until the marketplace is rebalanced

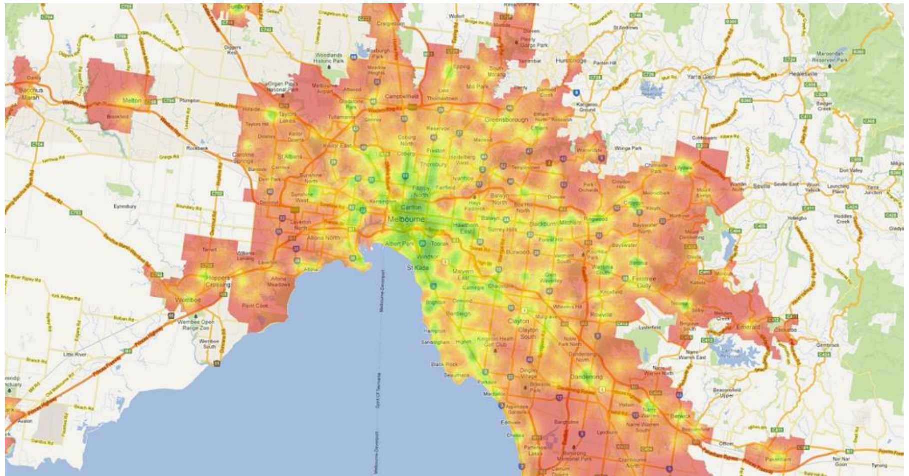


Supply and Demand – Econ101



Surge Pricing Mechanism

- Free market economy
  - Rising demand for rides
    - Fares goes up
  - Reduced demand for rides
    - Fares goes down
- Uber maps every city into granular hyperlocal zones
  - When demand increases in an area, the block will start changing the color
  - Price will be increased to (base rate x surge multiplier)
- Will this really work in real life?



Will that really work? - continued

- Cancelling the trip?
  - Surge is applied only to the basic fee – not the cancelation fee
  - Finding the rider’s battery level
- Will you pay for the Uber ride or take subway?
- Uber uses a lot of data to predict the “**willingness to pay**”
  - Combining various real-time data with user history with Uber
  - Event, holidays, weather, traffic, etc.
- World largest taxi company
  - Without a single taxi
  - More than 70 countries and over 10,500 cities worldwide.

Chen, M. Keith, and Michael Sheldon. "Dynamic pricing in a labor market: Surge pricing and flexible work on the Uber platform." Ec 16 (2016): 455.

What will be needed for the Uber technology?

- Microservices architecture
    - Thousands of independent single-purpose services communicating via remote procedure calls
- Polyglot persistence
    - Various data storage solution for schemaless massive data storage/retrieval
- Real-Time Dispatch System (DISCO)
    - Geospatial indexing (Hexagonal hierarchical spatial index, H3)
    - Supply& demand matching
        - Continuous push-based updates (every 4 seconds)
    - Routing engine – ETA prediction
    - Dynamic pricing
- …..

Course Introduction

Goals of CS435

- Understanding fundamental concepts and methodologies of Big Data
- Learn about effective Big Data technologies and how to apply them
  - Scalable computing paradigms
  - Scalable algorithms and models
  - Petascale data retrieval methodologies
  - End-to-end Big Data analytics with real-world datasets

Sangmi Lee Pallickara, Ph.D.

- Professor, Clare Boothe Luce Professor, Department of Computer Science, Colorado State University
- Big Data, GeoAI (Geospatial Artificial Intelligence), AI for Sciences, Distributed Geospatial Storage Systems, Data visualization
  - Over 100 scientific articles at journals and conferences
  - Advised 32 Masters thesis and 6 Ph.D. dissertations
- Teaching Big Data courses (CS435, 535), Data Mining at Scale (480)
  - Since 2011 at CSU
- Awards Highlights
  - National Science Foundation CAREER award
  - IEEE TCSC Award for Excellence in Scalable Computing
  - National Clare Boothe Luce Professorship
  - College of Natural Sciences' Jack Cochran Family Professorship

Seeking Undergrad Research Opportunities?

- Independent study/Honors thesis/research assistantship/undergrad research group
- *Are Parking Lots Cooking Fort Collins? - Correlation Analysis of Parking Lots and Heat Island Effects using Satellite Imagery and Open Street Maps*
    - 2nd place, Innovation in Energy and Climate Solutions, CSU's CURC, 2023
    - Junhwan Kim, Hermela Darebo, and Jackson Holden
- *Recommendation Guided Immersive Visual Exploration using Random Forests*
    - 2nd place, Data Science Research Track, CSU's CURC, 2022
    - Emma Hamilton, Mandey Brown, and Meridith McCann

If you are interested in CURC research poster team, please let me know. We will form a team in December 2026 for the CURC 2027 (April, 2027)!

Innovation in Energy and Climate Solutions: Second Place

Title: "Are Parking Lots Cooking Fort Collins? Correlation Analysis of Parking Lots and Heat Island Effects using Satellite Imagery and Open Street Maps"

Members: Junhwan Kim, Hermela Darebo and Jackson Holden

Faculty advisor: Sangmi Pallickara

Seeking Undergrad Research Opportunities?

We are currently looking for talented undergraduate research assistants. Please send me your resume!

XSD

PALLICKARA LAB

Course Communications

Canvas (Asynchronized communication)

- Primary course gateway
- Modules
- Announcements
- Grade
- Quizzes/worksheets (for online)
- Assignments

Meet the Team!

Sangmi Pallickara

- 9-9:50AM on Fridays, Office hours (CSB374 and Teams)
  - email me (Sangmi.Pallickara@colostate.edu) to arrange a separate sessions
- Lectures: 4-5:15PM on M/W, CSB130
- MeetYourProfessor (MYP) session for online section:(Sign up available in the Canvas announcement! You can arrange a separate meeting time as well.)

Course Components

LECTURES

Four Units
Unit 1: Introduction
Unit 2: In-memory distributed computing paradigm
Unit 3: Scalable Algorithms and Models
Unit 4: Petascale Data Storage and Retrieval
Worksheets (extra credits)
Recordings will be available via Canvas

Out-of-Class Activities

Two Programming Assignments
Term project
Five mandatory lab sessions
Help Desk hours (24hrs/week,
Workhours, evening, and weekends)

Assessments

Quizzes
Midterm and Final exam

Programming Assignments

- All assignments will be **individual** submissions
- Required programming language: **Java or Python**
- Grading will be based on **demo/interview in the CSB computing cluster**
- Late submission policy
  - Up to **2 days** of late submission is accepted
  - **10% deduction per-day** will be applied
- **Two Flex Tokens** for programming assignments
  - Each token will waive the late penalty of one day late submission
  - Minimum unit of token is “a day”
  - You can use two tokens together or use them separately
  - You must send an email to the GTA if you would like to use your flex token(s)
- Assignment submission
  - Canvas

Quizzes and Exams

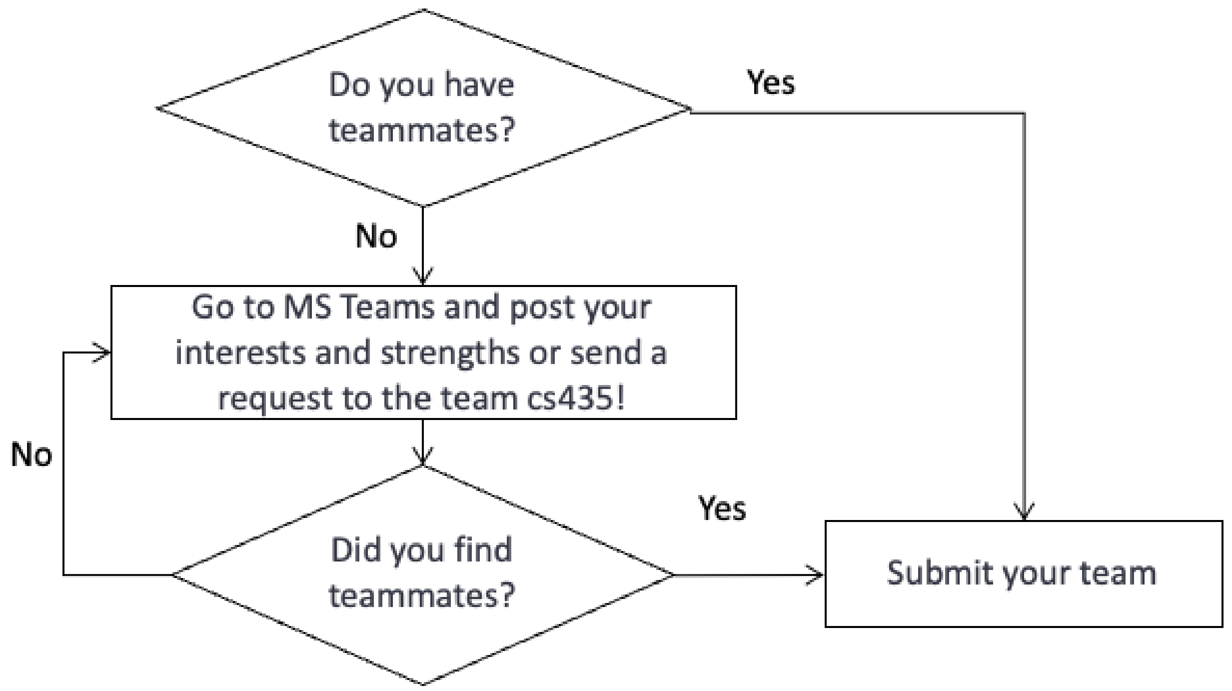
- About 10-15 quizzes this semester in class
  - Open slide, open book, No Internet, No collaboration
  - Online students will have online quizzes
  - Simple questions
  - **5% of final grade**
  - **3 lowest scores will be eliminated at the end of the semester**
- **No makeup quiz/worksheet provided**
- Midterm (25%) and Final (30%)
  - 55% of the course grade
  - Mixture of simple questions and comprehensive questions
  - Preparation guidelines will be provided

Term Project

- Comprehensive Big Data Analytics experiences
- Phase 0: Find your teammate!
- Phase 1: Term Project Proposal submission
- Phase 2: Final submission: Software and Report submission
- Phase 3: Presentation (Class presentation)

- Term project is a team effort **3-4 students per team**
  - No single member team or two-member team will be allowed!
  - Include one online student

How do I find my Term Project teammates?



tip Form a team with members who have "different" strengths and share "common" interests!

Example topics from previous courses

- Similarity and Clustering on The Million Song Dataset
- Methods Used to Analyze Eclipse and Mozilla Bug Data
- Movie Recommendations using Collaborative Filtering
- Climate Visualization and Predictive Analysis
- Wikipedia Page Traffic Statistics Analysis
  Trending Topics and Page Count Prediction on Wikipedia Traffic Log Data
- Supporting Emergency Response During Natural Disasters with Twitter Data
- Building a recommendation system for Spotify
- Regression Analysis of New York City Taxi Fares
- Analysis of crime patterns

Grading Policy

Category	Points	Percentage of Final Grade
Programming Assignments	PA1: 5% PA2: 5% 5 Lab sessions: 2% x 5 = 10%	20%
Term Project	Proposal: 1% Final report and software: 8% Discussion: 5% Team participation: 3% Final presentation: 3%	20 %
Quizzes	**5%**	5%
Exams (Midterm, Final)	**Midterm: 25** **Final: 30**	55%
Extra credits	**Weekly worksheets and participation**	5%
Total		105%

Final Letter Grade Scheme

Letter grades will be based on the following standard breakpoints

- >= 90 is an A,
- >= 89 is an A-,
- >=88 is a B+,
- >=80 is a B,
- >=79 is a B-,
- >=78 is a C+,
- >=70 is a C,
- >=60 is a D,
- and <60 is an F

Course Policy

- No make-up for missed exams
  - Except in extraordinary circumstances (e.g., major illness, family emergency) with CSU acceptable documents

- No make-up for missed quizzes/worksheets
  - 3 lowest scores will be eliminated
  - Except for the case where there is an emergency or University sanctioned events
  - Provide CSU acceptable documents: e.g., doctor’s note, CSU coach’s letter, etc.

Course Policy

- No textbook
  - Rely on the lecture notes
  - I provide several on-line materials and research papers
- Accommodation/honors thesis/honor requirements

Academic Integrity

- "*Plagiarism is the unauthorized or unacknowledged use of another person's academic or scholarly work. Done on purpose, it is cheating. Done accidentally, it is no less serious. Regardless of how it occurs, plagiarism is a theft of intellectual property and a violation of an ironclad rule demanding credit be given where credit is due.*"
- If you plagiarize in your work you could lose credit for the plagiarized submission, fail the assignment, or fail the course
- Use of Generative AI tools
  - Students may use generative AI tools for out-of-class activities (programming assignments and term project)
  - In this course, AI does NOT assume authorship or accountability
  - Transparency is required whenever AI makes a substantive contribution
  - e.g., add a chapter for your final term project report and PAs which list all the AI contributions in your submission

No laptop, hand-held device, or phone in class

- If you would like to use a laptop, please sit in the last row in the classroom
  - ATTENTION is OUR RESOURCE!
- **Check the Canvas modules, MS Teams discussion board at least twice a week**

Questions?